Anesthesia / OT Technician Questions (Part 5)

AIIMS CRE Exam Preparation Material based on Video Transcript

Q1. What is the internal diameter of the Endotracheal (ET) tube connector?

- 10 mm
- 12 mm
- 15 mm
- 18 mm

Correct Answer: 15 mm

Explanation: The internal diameter of the universal endotracheal tube connector, which connects the ET tube to the oxygen source/breathing circuit, is universally 15 mm.

Q2. What is the Minimum Alveolar Concentration (MAC) value of Sevoflurane?

- 0.25%
- 2%
- 6%
- 104%

Correct Answer: 2%

Explanation: The MAC value of Sevoflurane is approximately 2%. For comparison, Halothane is about 0.75%, Desflurane is 6%, and Nitrous Oxide is 104%.

Q3. Pin Index Safety System (PISS) is used for which gas system?

- Pipeline gas
- Liquid oxygen
- Medical air (pipeline)
- Cylinder gas

Correct Answer: Cylinder gas

Explanation: The Pin Index Safety System is a safety mechanism used specifically for small medical gas cylinders (like Type E cylinders attached to anesthesia machines) to prevent accidental connection of the wrong gas cylinder.

Q4. What is the color of the Nitrous Oxide (N2O) cylinder?

- White
- Blue
- Black
- Green

Correct Answer: Blue

Explanation: According to standard medical gas cylinder color coding, the Nitrous Oxide cylinder is painted blue. Oxygen cylinders have a black body with a white shoulder (in India) or green (in the US).

Q5. What is the color of the Air pipeline in an anesthesia workstation?

- Yellow
- Green
- Black
- Blue

Correct Answer: Black

Explanation: The color coding for central medical air pipelines attached to the anesthesia machine is black (with white/black markings depending on standards). Yellow is for vacuum/suction, blue is for N2O, and white is for Oxygen.

Q6. What is the normal cuff pressure of an Endotracheal Tube?

- 50 to 80 mmHg
- 80 to 100 mmHg
- 100 to 150 mmHg
- 25 to 30 cmH2O (approx 20-25 mmHg)

Correct Answer: 25 to 30 cmH2O (approx 20-25 mmHg)

Explanation: The normal ET tube cuff pressure should be maintained between 20 to 30 cmH2O (approx 15-22 mmHg) to prevent tracheal mucosal damage while ensuring an adequate seal to prevent aspiration.

Q7. What is the normal adult respiratory rate?

- 6 to 10 breaths per min
- 8 to 12 breaths per min
- 12 to 20 breaths per min
- 20 to 30 breaths per min

Correct Answer: 12 to 20 breaths per min

Explanation: The normal resting respiratory rate for a healthy adult is typically 12 to 20 breaths per minute.

Q8. What is the normal tidal volume in an adult?

- 200 to 300 ml
- 300 to 400 ml
- 400 to 500 ml
- 500 to 600 ml

Correct Answer: 500 to 600 ml

Explanation: The normal tidal volume for a healthy adult is approximately 500 ml per breath (roughly 6-8 ml/kg of ideal body weight).

Q9. What is the capacity of a Type-E Oxygen cylinder?

- 300 liters
- 400 liters
- 500 liters
- 660 liters

Correct Answer: 660 liters

Explanation: A standard E-type oxygen cylinder used on anesthesia machines holds approximately 660 liters of oxygen at a full pressure of 2000 psi.

Q10. Boyle's law works on which relationship between pressure and volume?

- Pressure is directly proportional to Volume
- Pressure is inversely proportional to Volume ($P \propto 1/V$)
- Volume is inversely proportional to Temperature
- Pressure is inversely proportional to Temperature

Correct Answer: Pressure is inversely proportional to Volume ($P \propto 1/V$)

Explanation: Boyle's Law states that at a constant temperature, the pressure of a given mass of an ideal gas is inversely proportional to its volume ($P \propto 1/V$).

Q11. A Laryngoscope blade is usually made of which material?

- Plastic
- Aluminum
- Stainless Steel
- Copper

Correct Answer: Stainless Steel

Explanation: Most reusable laryngoscope blades (and other surgical instruments) are made of stainless steel because it is durable, rust-resistant, and easy to sterilize.

Q12. Capnography is used to monitor:

- Oxygen Saturation
- Blood Pressure
- Heart Rate
- End-Tidal CO_2 ($EtCO_2$)

Correct Answer: End-Tidal CO_2 ($EtCO_2$)

Explanation: Capnography is the continuous non-invasive measurement and graphical display of the end-tidal carbon dioxide ($EtCO_2$) concentration during the respiratory cycle.

Q13. CO_2 absorber is used in which breathing system?

- Open system
- Semi-open system
- T-piece
- Closed system

Correct Answer: Closed system

Explanation: A CO_2 absorber (soda lime) is used in closed and semi-closed breathing circuits (circle systems) to chemically remove exhaled carbon dioxide, allowing the safe rebreathing of anesthetic gases.

Q14. Which is an ester type of local anesthesia agent?

- Lidocaine
- Bupivacaine
- Procaine
- Ropivacaine

Correct Answer: Procaine

Explanation: Procaine is an ester-type local anesthetic. A common mnemonic: amides have two 'i's in their generic name (e.g., LIdocaIne, bupIvacAIne), whereas esters have only one 'i' (e.g., procAInE, chloroprocAInE).